SOC Alert Monitoring & Log Correlation

Using Sysmon, Windows Event Viewer, Wireshark & a Python-Based SOC Incident Analyzer

Undergraduate Cybersecurity Capstone Project

Agenda

- Project Overview
- Lab Setup & Architecture
- Data Collection
- Attack Simulation
- Log Correlation & Timeline
- Findings & Alerts
- MITRE ATT&CK Mapping
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Project Overview

- SOC Alert Monitoring & Log Correlation exercise
- Virtualized lab: Windows 11 host + Kali Linux attacker (VMware Workstation Pro)
- Telemetry sources: Sysmon, Windows Event Viewer, Wireshark
- Custom Python SOC Incident Analyzer (v3.0) for correlation & reporting
- Result: 9 incidents identified, Risk Score 69/100 (HIGH)

Lab Setup & Architecture

- Platform: VMware Workstation Pro
- Monitored endpoint: Windows 11 (victim host)
- Attacking machine: Kali Linux 2026.1
- Isolated internal networking via VMware VMnet adapters
- Host IP address: 192.168.0.105 (Wi-Fi adapter)
- No exposure to external network

Data Collection

- Sysmon: Event ID 1 (Process Creation)
- Sysmon: Event ID 3 (Network Connection)
- Windows Event Viewer: Event ID 4624 (Successful Logon)
- Windows Event Viewer: Event ID 4625 (Failed Logon)
- Wireshark: Packet capture, filter tcp.port == 135
- Captured TCP 3-way handshake (SYN, SYN-ACK, ACK) & RST/ACK resets

Attack Simulation

- Attacking machine: Kali Linux
- Tool: Nmap TCP connect scan (nmap -Pn -sT)
- Target: Windows 11 host (192.168.0.105)
- Open ports found: 135 (msrpc), 139 (netbios-ssn), 445 (microsoft-ds)
- Additional ports observed in earlier scans: 808, 9001
- Multiple scans run to confirm consistency

Log Correlation & Timeline

- Engine: Python SOC Incident Analyzer (v3.0)
- Log entries processed: 17
- Correlated sources: Sysmon + Event Viewer + Wireshark
- Built-in Attack Timeline correlation function
- Output: 9 incidents, 13 Indicators of Compromise (IOCs)
- Top indicators: powershell.exe, Administrator account

Findings & Alerts

- Overall Risk Score: 69 / 100 (HIGH)
- Severity: 2 Critical, 4 High, 1 Medium, 2 Low
- Detection rate reported by tool: 100%
- Critical incident INC-0013: PowerShell process → external C2 callback
- Indicators of Compromise (IOCs) extracted: 16
- Categories: Authentication Success, Credential Access, Network Recon, Process Creation

MITRE ATT&CK Mapping

- T1046 – Network Service Discovery (Nmap recon)
- T1059 – Command and Scripting Interpreter
- T1071.001 – Application Layer Protocol (C2 Callback)
- T1110 – Brute Force
- T1547 – Registry Run Keys / Startup Folder
- Critical: PowerShell C2 callback (INC-0013) mapped to T1071.001

Conclusion

- Simulated end-to-end SOC analyst workflow
- Correlated telemetry from Sysmon, Event Viewer & Wireshark
- 9 incidents identified; overall HIGH risk rating (69/100)
- Python SOC Incident Analyzer automated classification, scoring, MITRE mapping & reporting
- Future work: real-time log ingestion, firewall/proxy logs, wider attack simulation

Questions ?

Thank You!